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Inventor(s)

Solheid; James J. et al.

TELECOMMUNICATIONS MODULE AND FRAME

Abstract

A telecommunications chassis comprises a cable sealing portion defining at least one cable opening configured to sealably receive a cable and a module mounting portion extending from the cable sealing portion, which further comprises a housing defining an open front closable by a door to define an interior, a rear wall, a right wall, and a left wall. A plurality of module mounting locations is provided in a vertically stacked arrangement, each configured to receive a telecommunications module through the open front. An exterior of the housing includes a first column of radius limiters defining curved profiles for guiding cables from the front toward the rear with bend control. A second column of radius limiters in the form of spools is spaced apart and generally parallel to the first column of radius limiters and a third column of radius limiters, at least some of which are in the form of spools, is also spaced apart and generally parallel to the first and second columns of radius limiters. The rear wall defines an opening for accessing from the exterior of the housing rear ends of modules to be mounted in the housing for signal input, wherein the exterior also includes a plate at least partially overlapping the opening for protection of cables entering the opening.

Inventors: Solheid; James J. (Minneapolis, MN), Bolster; Kristofer (Jordan, MN), Vongseng; Soutsada (Chanhassen, MN), LeBlanc; Thomas G. (Westminster, MA)

Applicant: CommScope Technologies LLC (Claremont, NC)

Family ID: 59998064

Assignee: CommScope Technologies LLC (Claremont, NC)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/607,987, filed Mar. 18, 2024; which is a continuation of U.S. application Ser. No. 17/936,185, filed Sep. 28, 2022, now U.S. Pat. No. 11,934,024; which is a continuation of U.S. application Ser. No. 17/065,726, filed Oct. 8, 2020, now U.S. Pat. No. 11,467,362; which is a continuation of U.S. application Ser. No. 16/290,512, filed Mar. 1, 2019, now U.S. Pat. No. 10,802,238; which is a continuation of U.S. application Ser. No. 15/479,463, filed Apr. 5, 2017, now U.S. Pat. No. 10,222,571; which claims the benefit of U.S. Provisional Application No. 62/319,535, filed Apr. 7, 2016, the disclosures of which is hereby incorporated herein by reference.

FIELD

[0002] The present disclosure generally relates to fiber optic telecommunications equipment. More specifically, the present disclosure relates to modular elements housing fiber optic telecommunications equipment that are mounted to telecommunications frames.

BACKGROUND

[0003] In fiber optic telecommunications systems, it is common for optical fibers of transmission cables to be split into multiple strands, either by optical splitting of a signal carried by a single stranded cable or by fanning out the individual fibers of a multi-strand cable. Further, when such systems are installed, it is known to provide excess capacity in the installations to support future growth and utilization of the fibers. Often in these installations, modules including splitters or fanouts are used to provide the connection between transmission fibers and customer fibers. To reduce the cost and complexity of the initial installation and still provide options for future expansion, a module mounting fixture such as a chassis or a tray capable of mounting multiple modules may be used in such an installation.

[0004] While the fixture may accept several modules, the initial installation may only include fewer modules mounted in the fixture, or enough to serve current needs. These fixtures may be configured with limited access to one or more sides, or may be mounted in cramped locations. In addition, some of these fixtures may be pre-configured with the maximum capacity of transmission cables to accommodate and link to modules which may be installed in the future. Since it is

desirable to have access to components within the fixture for cleaning during the installation of a new module, some provision or feature of the fixture will desirably permit a user to access and clean the connectors of these pre-connectorized and pre-installed transmission cables.

[0005] In fiber-optic telecommunications, it is also common for optical signals of transmission cables to be multiplexed. Wavelength division multiplexing (WDM) is a technology which multiplexes multiple optical carrier signals on a single optical fiber by using different wavelengths of laser light to carry different signals. This allows for a multiplication in capacity, in addition to making it possible to perform bidirectional communications over one strand of fiber.

[0006] Improvements in the design of such telecommunications modules and the telecommunications frames for mounting them are desired.

SUMMARY

[0007] The present disclosure relates to a telecommunications assembly including a telecommunications fixture such as a tray and at least one telecommunications module mounted within the tray. Within an interior of each of the modules is located a fiber optic component. In one embodiment, the fiber optic component may be a fiber optic splitter. In another embodiment, the fiber optic component may be a fiber optic division multiplexer/demultiplexer. The module may include one or more signal input locations and one or more signal output locations. In certain embodiments, the signal input locations may be adjacent the signal output locations, and the cabling extending from the signal input locations to the fiber optic component and then to the signal output locations from the fiber optic component may utilize the same cable management features within the module housing. When the module is used as a fiber optic division multiplexer/demultiplexer module, the multiplexer/demultiplexer, as a receiver, is configured to demultiplex multiple optical carrier signals carried by the single input optical fiber into different wavelengths of laserlight as customer output signals. As a transmitter, the multiplexer/demultiplexer is configured to multiplex the customer signals, which are different wavelengths of laserlight, and combine them into a single optical fiber to be outputted from the module.

[0008] According to another aspect of the present disclosure, the module comprises a housing including a main housing portion defining an interior formed by a first sidewall, a second sidewall, a bottom wall, a front wall, and a rear wall, the housing further including a removable cover mounted to the main housing portion to define a top wall. The interior defines a right chamber separated from a left chamber. A fiber optic component is housed within the left chamber. A signal input location for receiving an input signal to be processed by the fiber optic component and a signal output location for exiting an output signal processed by the fiber optic component are both exposed to the right chamber, wherein both the signal input location and the signal output location are defined by the front wall, wherein a cable carrying the input signal is fixed adjacent the signal input location via a crimp assembly, and a cable carrying the output signal is fixed adjacent the signal output location via a crimp assembly, wherein an excess fiber length formed when an outer jacket of the cables contracts more than the fiber therewithin due to temperature variations is accommodated by the right chamber to allow the excess fiber length to accumulate without bending in a radius smaller than a minimum bend radius. A cable management structure is positioned within the right chamber, the cable management structure being a dual-layered cable management structure defining a lower cable-wrapping level and a separate upper cable-wrapping level, wherein the upper cable-wrapping level is defined by a removable cable retainer that is mounted on a spool defining the lower-cable wrapping level, the dual layered cable management structure including both the cable carrying the input signal and the cable carrying the output signal wrapped therearound for cable management. The cable carrying the input signal and the cable carrying the output signal are passed between the right chamber and the left chamber before and after being processed by the fiber optic component, respectively, wherein at least one of the rear wall, the first sidewall, and the second sidewall defines a curved portion for providing bend radius protection to

the cables carrying the input and output signals within the module.

[0009] According to another aspect, the disclosure is directed to a telecommunications chassis/frame for receiving a plurality of the above-discussed modules.

[0010] According to one inventive aspect, such a chassis includes a cable sealing portion defining at least one cable opening configured to sealably receive a cable and a module mounting portion extending from the cable sealing portion. The module mounting portion further comprises a housing defining an open front closable by a door to define an interior, a rear wall, a right wall, and a left wall, the housing further defining a plurality of module mounting locations provided in a vertically stacked arrangement within the interior, each configured to receive a telecommunications module through the open front, an exterior of the housing including a first column of radius limiters each defining a curved profile for guiding cables from the front of the housing toward the rear with bend control, the exterior of the housing including a second column of radius limiters in the form of spools that are spaced apart and generally parallel to the first column of radius limiters, the exterior of the housing further including a third column of radius limiters, at least some of which are in the form of spools that are spaced apart and generally parallel to the first and second columns of radius limiters, the rear wall defining an opening for accessing from the exterior of the housing rear ends of modules to be mounted in the housing for signal input, wherein the exterior of the housing includes a plate at least partially overlapping the opening for protection of cables entering the opening.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The accompanying drawings, which are incorporated in and constitute a part of the description, illustrate several aspects of the inventive features and together with the detailed description, serve to explain the principles of the disclosure. A brief description of the drawings is as follows.

[0012] FIG. 1 is a front perspective view of a telecommunications assembly including a telecommunications module having features that are examples of inventive aspects in accordance with the present disclosure mounted within a telecommunications fixture in the form of a tray.

[0013] FIG. 2 illustrates the telecommunications assembly of FIG. 1 from a top view.

[0014] FIG. 3 illustrates the telecommunications assembly of FIG. 1 with the module exploded off the tray.

[0015] FIG. 4 illustrates the telecommunications assembly of FIG. 1 with the cover of the module removed from the main housing portion of the module housing.

[0016] FIG. 5 illustrates the main housing portion of the module housing of FIG. 4 with a number of features exploded from the main housing portion, the module shown without any optical components or associated cabling therein.

[0017] FIG. 5A is a close-up view of the cable exit portion of the module of FIG. 5.

[0018] FIG. 6 illustrates the main housing portion of FIG. 5 from a top view wherein all of the features thereof are in an assembled configuration.

[0019] FIG. 7 illustrates the main housing portion of FIG. 5 in a fully populated configuration with the optical component and the cabling therein.

[0020] FIG. 8 is cross-sectional view taken along line 8-8 of FIG. 7.

[0021] FIG. 8A is a close-up view of the output cable crimp holding portion of the module of FIG. 8.

[0022] FIG. 9 illustrates a front view of the insect-infestation prevention device of the module of FIG. 8 in isolation.

[0023] FIGS. 10-13 illustrate the sequential steps for routing cabling associated with an input

signal that is split by a fiber optic splitter of the module into a plurality of output signals output from the module.

[0024] FIG. **14** illustrates an exploded view of an example of a crimp assembly (i.e., furcation tube assembly) that can be used with the output pigtails of the module of FIGS. **1-13**.

[0025] FIG. **15** illustrates the crimp assembly of FIG. **14** in an assembled configuration.

[0026] FIG. **16** illustrates an exploded view of an example of a crimp assembly (i.e., furcation tube assembly) that can be used with the input cabling of the module of FIGS. **1-13**.

[0027] FIG. **17** illustrates the crimp assembly of FIG. **16** in an assembled configuration.

[0028] FIG. **18** is a front perspective view of a telecommunications chassis/frame configured to receive a plurality of the telecommunications assemblies of FIGS. **1-13**, the chassis shown populated with a plurality of telecommunications assemblies and shown in an open access position.

[0029] FIG. **19** is a front view of the telecommunications chassis/frame of FIG. **18**.

[0030] FIG. **20** is a rear perspective view of the telecommunications chassis/frame of FIG. **18**.

[0031] FIG. **21** illustrates the chassis/frame of FIG. **18** in an empty configuration.

[0032] FIG. **22** illustrates the chassis/frame of FIG. **19** in an empty configuration.

[0033] FIG. **23** illustrates the chassis/frame of FIG. **20** in an empty configuration.

[0034] FIG. **24** illustrates a partially exploded view of a sealed enclosure formed by clamping a cover/dome on the chassis/frame of FIGS. **18-23**.

[0035] FIG. **25** illustrates the enclosure of FIG. **24** in an assembled configuration.

[0036] FIG. **26** illustrates a partially exploded view of the enclosure of FIGS. **24-25** being mounted on a vertical surface such as a telecommunications pole.

[0037] FIG. **27** illustrates the enclosure of FIG. **26** in a fully mounted position.

[0038] FIG. **28** is a front perspective view of another embodiment of a telecommunications chassis/frame configured to receive a plurality of the telecommunications assemblies of FIGS. **1-13**, the chassis shown populated with a plurality of telecommunications assemblies and shown in an open access position.

[0039] FIG. **29** is a front view of the telecommunications chassis/frame of FIG. **28**.

[0040] FIG. **30** is a rear perspective view of the telecommunications chassis/frame of FIG. **28**.

[0041] FIG. **31** illustrates the chassis/frame of FIG. **28** in an empty configuration.

[0042] FIG. **32** illustrates the chassis/frame of FIG. **29** in an empty configuration.

[0043] FIG. **33** illustrates the chassis/frame of FIG. **30** in an empty configuration.

[0044] FIG. **34** illustrates a partially exploded view of a sealed enclosure formed by clamping a cover/dome on the chassis/frame of FIGS. **28-33**.

[0045] FIG. **35** illustrates the enclosure of FIG. **34** in an assembled configuration.

[0046] FIG. **36** illustrates a partially exploded view of the enclosure of FIGS. **34-35** being mounted on a vertical surface such as a telecommunications pole.

[0047] FIG. **37** illustrates the enclosure of FIG. **36** in a fully mounted position.

DETAILED DESCRIPTION

[0048] Reference will now be made in detail to exemplary aspects of the present disclosure which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or similar parts.

[0049] FIGS. **1-4** illustrate a telecommunications assembly **10** including a telecommunications module **12** having features that are examples of inventive aspects in accordance with the present disclosure mounted within a telecommunications fixture in the form of a tray **14**. The depicted fixture in the form of a tray **14** may be mounted to other telecommunications fixtures such as chassis/frame, etc.

[0050] In the given embodiment, the telecommunications module **12** defines notches **16** on a right sidewall **18** and a left sidewall **20** of a main housing portion **22** of the module housing **24** for receiving elastic ramped tabs **26** provided on the tray **14** for mounting the module **12** to the tray **14** with a snap-fit interlock.

[0051] A bottom wall **28** defined by the module main housing portion **22** also includes a cutout **30** extending in a front-to-back direction for accommodating a center divider **32** of the tray **14**. The center divider **32** of the tray **14** extends all the way across the tray **14** and transitions into a cable management finger **34** that protrudes outwardly from the front of the tray **14**.

[0052] It should be noted that although the module **12** of the present disclosure is depicted as being mounted on a telecommunications tray **14**, the module **12** may be utilized in other types of fixtures, and the depicted tray **14** is only one example of such a fixture.

[0053] FIGS. **5-13** illustrate the telecommunications module **12** having features that are examples of inventive aspects in accordance with the present disclosure. Since one of the telecommunications equipment housed by the module is a fiber optic splitter **36**, the telecommunications module **12** may also be referred to herein as a fiber optic splitter module **12**. It should be noted that an optical splitter **36** is only one example of an optical component that may be housed within the module **12** of the present disclosure, and in other embodiments, the module **12** may be configured to house other types of optical equipment such as fan-outs, wavelength division multiplexer/demultiplexers, combiners, filters, etc. As will be discussed in further detail below, according to certain embodiments, the module **12** may be provided with removable inserts that are customized based on the type of fiber optic equipment that is going to be housed within the module **12** and the corresponding cable routing provided within the module **12**. The inserts may be sized and configured to accommodate the different types of fiber optic equipment and may include pre-mounted or pre-molded cable management structures such as radius limiters or spools that are designed in accordance with the cable routing needed for the types of fiber optic equipment provided in the module **12**.

[0054] The telecommunications module **12** includes the module housing **24** that is made up of the main housing portion **22** and a removable cover **38** (shown in an exploded configuration in FIG. **4**). The cover **38** may be mounted to the module main housing portion **22** via fasteners **40** as shown in FIGS. **4** and **5**.

[0055] In FIGS. **5** and **5A**, the main housing portion **22** of the module housing **24** is shown in an exploded orientation.

[0056] The main housing portion **22** and the removable cover **38** of the module **12** are illustrated in a fully assembled configuration in FIGS. **1-3**. The main housing portion **22** of the module housing **24** is shown without the optical equipment and associated cabling in FIGS. **5**, **5A**, and **6**. In FIGS. **7** and **8**, the module **12** is shown fully populated with these features.

[0057] Still referring to FIGS. **5-13**, in the given embodiment, the module **12** is depicted as being configured to house an optical component in the form of a fiber optic splitter **36**. The fiber optic splitter **36** is adapted to power split a first input fiber optic signal entering the module **12** into multiple strands that are output as connectorized pigtails **42** from the module **12**. In the embodiment shown, the fiber optic splitter **36** is a 1×32 splitter. As such, the module **12** of the present disclosure is illustrated as outputting **32** output signals via fiber optic pigtails **42** for every input signal entering the module **12**. As shown and as will be discussed in further detail, an outer layer (e.g., a strength layer **44**) of the cabling **46** carrying the input signal, and an outer layer (e.g., a strength layer **48**) of the cabling (e.g., pigtails) **42** carrying the output signals may be fixed to the module housing **24** via crimping so as to still allow the internal fibers to move through the walls of the module housing **24** for accommodating thermal expansion.

[0058] Now referring specifically to FIGS. **5**, **5A**, and **6**, the main housing portion **22** of the housing **24** of the fiber optic splitter module **12** is illustrated. The cover **38** of the fiber optic splitter module housing **24** has been removed, exposing the interior features of the fiber optic splitter module **12** including an example cable routing associated with a splitter **36** within the fiber optic splitter module **12**.

[0059] The main housing portion **22** defines the bottom wall **28** extending between the right wall **18**, the left wall **20**, a rear wall **52**, and a front wall **54**. As noted above, the open side **56** of module

main housing **22** is normally closed by a removable cover **38** that defines a top wall **58** of the module housing **24**.

[0060] Rear wall **52** of main housing portion **22** includes curved portions **60** that are configured to provide bend radius protection and routing guidance to cables **42**, **46** within an interior **62** of the module **12**. The front wall **54** of module main housing portion **22** is configured to provide the signal input and the signal output locations **64**, **66** of the module **12**. The depicted module **12** includes two signal input openings **68** at the front wall **54**. The signal input openings **68** of the front wall **54** are normally covered by downwardly protruding tabs **70** of the cover **38**. The protruding tabs **70** may include breakable portions **72** for exposing one or more of the signal input openings **68** for utilizing the opening as a signal input location **64** (please refer to FIG. 4).

[0061] The depicted module **12** also defines a signal output or a cable exit opening **74** at the front wall **54**, adjacent the signal input openings **68**. The module **12** is configured to receive a bend limiting apparatus **76** at the signal output opening **74** that extends outwardly from the front wall **54**. The bend limiting apparatus **76** is in the form of a boot **50** and is configured to provide bend protection to the plurality of pigtails **42** carrying the output signals exiting the module **12**.

[0062] The boot **50** defines a central opening **78** aligned with the cable exit opening **74** for routing the pigtails **42** out of the module **12**. The boot **50** is slidably mounted to main housing portion **22** and is configured to be captured by the cover **38** of module **12** when cover **38** is mounted to main housing portion **22**.

[0063] According to the depicted embodiment, in order to facilitate placement and/or removal of the flexible boot **50**, the cable exit opening **74** defined by the main housing portion **22** is configured such that the boot **50** can be axially slid over the pigtails **42**, brought toward the module **12** from an exterior of the module **12**, and placed at the cable exit opening **74**, wherein a cable passage defined by the central opening **78** of the boot **50** aligns with the cable exit opening **74**. In the depicted embodiment, a rear lip or flange **80** of the boot **50** can be inserted through the cable opening **74** and then slidably placed within a first notch **82** defined at the cable opening **74**. Once the rear lip **80** of the boot **50** has been placed within the first notch **82**, a separate boot retainer **84** in the form of a C-shaped clip is slidably placed over the boot **50** in a direction transverse to the axial direction. The boot retainer **84** defines a flange **86** that is slidably placed within a second notch **87** defined in front of the first notch **82**. Once positioned, the flange **86** defined by the boot retainer **84** abuts, on one side, an inner side of the front wall **54** surrounding the cable exit opening **74**. And, the flange **86** defined by the boot retainer **84** also abuts, on the opposite side, the lip **80** of the boot **50** to prevent the boot **50** from being pulled out in the axial direction. The cover **38** of the module housing **24** is placed on the main housing portion **22** and captures the boot retainer **84** against the main housing portion **22** to prevent the boot **50** from being pulled out.

[0064] According to the depicted embodiment, the fiber optic splitter module also includes at least one insect-infestation prevention device **88** that is configured to be placed at the cable exit opening **74**. The insect-infestation prevent device **88** is shown in isolation in FIG. 9. The cover **38** of the module **12** includes structure for capturing the insect-infestation prevention device **88** within the main housing portion **22** of the fiber optic splitter module housing **24**, as will be described in further detail.

[0065] The insect-infestation prevention device **88** is configured to be mounted adjacent the front cable exit opening **74** of the main housing portion **22**, aligning with the boot central opening **78**. As shown, the insect-infestation prevention device **88** includes a one-piece molded body **90** defining a top end **92**, a bottom end **94**, a front end **96**, a rear end **98**, a right side **100** and a left side **102**. The body **90** includes cutouts **104** extending from the top end **92** toward the bottom end **94**. The cutouts **104** are configured to receive fiber optic cables in a direction from the top end **92** toward the bottom end **94**. In the depicted embodiment, the cutouts **104** each include a zig-zag configuration (a.k.a., an S-shaped configuration). The zig-zag configuration is designed to increase the density of the cables that can be stacked in the cutouts **104**. For example, in the depicted embodiment, the zig-

zag configuration allows a cutout **104** to accommodate four fiber optic cables stacked on top of each other, whereas straight slots would require a wider footprint for the cable manager or deeper cutouts. In other embodiments, depending upon the density required, the cutouts **104** could have straight slotted configurations.

[0066] The angles forming the zig-zag configuration of the cutouts **104** are designed such that, while providing a higher density for the cables, they also preserve bend radius properties of the fibers. When the insect-infestation prevention device **88** is aligned with the boot **50**, the cutouts **104** are preferably sized to provide a snug fit with the output cables **42** such that any extra room around the cables **42** is limited to prevent insect-infestation. When less than all of the cutouts **104** or less than the entirety of a cutout **104** is used, the remaining space can be plugged using different types of inserts. For example, according to one embodiment, the remaining space that is not used for outputting a fiber can be plugged using a dummy fiber furcation tube assembly. The dummy fiber furcation tube assembly may include all of the components of a normal live fiber furcation tube assembly except for the fiber itself. An example of a dummy fiber furcation tube assembly is discussed in U.S. Pat. No. 8,824,850, the entire disclosure of which is incorporated herein by reference.

[0067] The rear end **98** of the body **90** of the insect-infestation prevention device **88** includes right and left flanges **106**, **108**. As shown in FIGS. 5, 5A, and 6, the insect-infestation prevention device **88** is slidably mounted at the cable exit opening **74**, behind the boot **50**, such that the right and left flanges **106**, **108** are inserted into a third notch **110** defined adjacent the cable exit opening **74**. Once the flanges **106**, **108** are within the third notch **110**, the front end **96** of the insect-infestation prevention device **88** lies flush with the rear end of the boot **50**.

[0068] Once the insect-infestation prevention device **88**, the boot **50**, and the boot retainer **84** are placed on the main housing portion **22**, the cover **38** is used to capture these components against the main housing portion **22**.

[0069] As discussed previously, an outer layer **48** of the cables **42** exiting the module **12** through the boot **50** is fixed to the main housing portion **22** of the module **12** so as to allow expansion of the fiber through the cabling **42** due to thermal variances.

[0070] According to the depicted embodiment, the module **12** includes integrally formed crimp holders **112** (e.g., slots) within the interior **62** of the module housing **24** adjacent the front wall **54** thereof. Crimp elements **114** (see FIGS. 14 and 15) crimped to the output pigtail cables **42** are slidably received into the crimp holders **112**. Crimp elements **114** include square flanges **116** between which are defined recessed portions **118**. The crimp holders **112** include complementary structure to the crimp elements **114** such that once the crimp elements **114** are slidably inserted into the crimp holders **112**, the crimp elements **114** are prevented from moving in a longitudinal direction due to the flanges **116**. Once slidably inserted, crimp elements **114** are held in place by the cover **38** that is mounted on the module main housing **22**. The crimp assembly **120** used for the output cables **42** is illustrated in FIGS. 14 and 15 and discussed in further detail below.

[0071] In the embodiment shown, there are seven crimp holding slots **112**, each slot **112** being able to accommodate up to five crimp elements **114** (see FIGS. 8 and 8A). In the embodiment of the module **12** shown, since a 1×32 fiber optic splitter **36** may be used, the crimp holders **112** provide the capacity for all of the thirty-two crimp elements **114** connected to output cables **42**. Since only thirty-two crimp holding positions are occupied, the remaining three positions may be filled with dummy crimp elements **122** or inserts/fillers that are not connected to cables, making sure the crimp elements **114** crimped to active output cables **42** do not slide out of the slots **112**.

[0072] The configuration of the module housing **24** can certainly be modified to accommodate other number of inputs or outputs, as desired. In addition, other complementary shapes between the crimp elements **114**, and the crimp holders **112** can be used to provide a slidable fit and to prevent axial movement of the crimp elements **114** within the crimp holders **112**.

[0073] Referring now to FIGS. 14 and 15, one example of a crimp assembly **120** (i.e., furcation

tube assembly) that can be used with the module **12** is illustrated. The crimp assembly **120** includes, as discussed above, a crimp element **114** that defines a front portion with a circumferential notch **118** (i.e., recessed portion) configured to slidably fit within one of the crimp slots **112** of the module **12** and a rear portion **124** configured to crimp a strength member **48** of the cabling **42**. The crimp element **114** defines a central hollow opening through which an inner fiber bearing tube **126** extends. A strength member **48** is shown in FIG. **14** extending between the inner tube **126** and the outer tubing or jacket **128** of the cable **42**. Strength member **48**, as shown, is an aramid fiber such as Kevlar® but other suitable materials may also be used. Before crimping the strength member **48** to the crimp element **114**, the strength member **48** is overlapped onto rear portion **124** of the crimp element **114**. A crimp sleeve **130** is fit about the strength member **48** above the rear portion **124** of the crimp element **114** and is crimped to hold these elements together. The outer jacket **128** abuts against the rear end of the crimp sleeve **130**.

[0074] A similar crimp-based fixation assembly **131** is also provided for the signal input location **64**. As shown in FIGS. **16** and **17**, a strain relief boot **132** is provided that mates with a crimp element **134** for fixing the cabling **46** at the input location **64**. The crimp element **134** defines a circumferential notch **136** (i.e., recessed portion). The circumferential notch **136** is slidably inserted into one of the signal input openings **68** defined on the front wall **54** of the main housing portion **22**. The crimp element **134** of the input connection is captured by the cover **38** when the cover **38** is mounted on the main housing portion **22**.

[0075] The crimp assembly **131** at the signal input location **64** further includes an insertion cap **138** that mounts inside a front end of the crimp element **134** and a crimp sleeve **140** that mounts around the exterior of the front end of the crimp element **134**. The crimp element **134** is configured to receive, anchor, and provide strain relief/bend radius protection to the fiber optic cable **46** carrying the input signal. The input fiber optic cable **46** includes a jacket **142** surrounding a fiber bearing tube **144**. The fiber optic cable **46** also includes the strength layer **44** formed by a plurality of strength members (e.g., reinforcing fibers such as aramid yarn/Kevlar®) positioned between the optical fiber tube **144** and the outer jacket **142**. An end portion of the strength layer **44** is crimped between the crimp sleeve **140** and the exterior surface of the front end of the crimp element **134** so as to anchor the strength layer **44** to the crimp element **134**. The crimp assembly **131** further includes the strain relief boot **132** mounted at the front end of the crimp element **134**, over the crimp sleeve **140**, for providing strain relief and bend radius protection to the optical fiber.

[0076] The insertion cap **138** reduces the size of the central opening of the crimp element **134** such that the outer tubing or jacket **142** of the cable **46** can abut against the cap **138** as the inner tubing **144** bearing the fiber is inserted through the crimp element **134**.

[0077] Thus, the cable **46** carrying the input signal and the pigtails **42** carrying the output signals are both fixed to the module housing **24** while the fibers extending therethrough are allowed pass through the housing walls and expand and contract due to thermal variations. The module housing **24**, thus, provides an expansion chamber effect for the incoming and outgoing fibers.

[0078] Now referring back to FIGS. **5-13**, the interior **62** of the main housing portion **22** is divided into two halves or chambers **146**, **148** (e.g., a right chamber **146** and a left chamber **148**) by a divider wall **150**. The right chamber **146** may be considered the cable management and expansion chamber, and the left chamber **148** may be considered the optical equipment or component holding chamber.

[0079] Toward the rear wall **52** within the right chamber **146** is positioned a cable management structure in the form of a spool **152** that is integrally formed with the bottom wall **28** of the main housing portion **22**. A removably mounted cable retainer **154** is placed over the spool **152** so as to form a dual-layered cable management structure **156** as will be discussed in further detail below.

[0080] The cable retainer **154** defines circumferentially placed fingers **158** for retaining the cables around the dual-layered structure **156**. When the cable retainer **154** is placed on the spool **152** that is integrally formed with the bottom wall **28** of the main housing portion **22**, the fingers **158** fit

within notches **160** on the spool **152** to create a dual-layered structure.

[0081] The left chamber **148** of the main housing portion **22** is configured to house optical equipment or components for the module **12**. As noted above, the depicted module **12** is a fiber optic splitter module and a power splitter **36** is depicted as being housed within the left chamber **148**.

[0082] It should be noted that the left chamber **148** provides enough spacing to directly mount different types of optical equipment, or, alternatively, receive molded inserts that may have features for mounting different types of optical equipment. In the embodiment shown, the fiber optic splitter **36** is shown as being directly placed within the left chamber **148**.

[0083] Referring to FIGS. **7**, **8**, and **10-13**, an example of a cable routing path going all the way from the signal input location **64** to the signal output location **66** will now be described, referring to the features of the module **12** that allow management or provide for bend radius protection for such cabling.

[0084] FIGS. **7** and **8** illustrate a fully populated module **12** showing all of the cabling associated with the input and output signals. FIGS. **10-13** illustrate the example routing in sequential steps to facilitate understanding of the routing within the module **12**.

[0085] Now referring to FIG. **10**, in the depicted example, a cable **46** carrying the input signal enters the module housing **24** through the signal input opening **68** at the front wall **54** of the main housing portion **22**. That cable **46** is wrapped around the dual layer cable management structure **156** twice, with the final wrap ending up on the upper level of the cable management structure **156**. The initial wrap of the cable **46** is around the lower portion of the cable management structure **156** that is integrally formed with the bottom wall **28** of the module housing **24**, and this wrap allows expansion of the cable within the right chamber **146** of the module housing **24**. When the cable **46** passes from the lower level to the upper level of the cable management structure **156**, it is routed through a pair of cable holders **162** that limit movement of the cable **46** that is at the upper level of the cable management structure **156**.

[0086] From the upper level of the cable management structure **156**, the cable **46** passes over a notch **163** on the central divider **150** to the left chamber **148** of the module housing **24**. After passing to the left chamber **148** of the module housing **24**, the cable **46** is routed to follow an S-shaped path and initially extends all the way around the fiber optic splitter **36**, through three other cable holders **162** (one being located toward the front and left corner of the module housing **24**, the second being located adjacent the rear wall **52** of the housing **24**, and the third being located adjacent the divider **150**) before being spliced to the splice input side **164** of a splice area **166**.

[0087] Referring now to FIG. **11**, from the splice output side **168** of the splice area **166**, the cable **46** carrying the input signal is then routed again all the way around the fiber optic splitter **36** in the left chamber **148**, passing through the three cable holders **162**, before entering the splitter input side **170** of the fiber optic splitter **36**.

[0088] FIG. **12** illustrates the cable routing for the cables **42** that are output from the splitter output side **172** of the fiber optic splitter **36**. As shown in FIG. **12**, after the cables **42** carrying the output signal leave the output side **172** of the splitter **36**, the cables **42** are routed all the way around the left chamber **148**, passing through the three cable holders **162**, before being led to the splice input side **164** of the splice area **166**.

[0089] Now referring to FIG. **13**, after being spliced to cabling **42** carrying the output signal, the cables **42** are lead from the splice output side **168**, all the way around the left chamber **148** again, toward the right chamber **146** of the module housing **24**. The cables **42** pass over the divider **150** and are then routed around the cable management structure **156** before being led to the crimp holders **112**. As discussed previously, the cables **42** carrying the output signal are crimped via crimp elements **114**, and the strength members **48** thereof are fixed to the module housing **24** at the crimp area, before being led out as pigtails **42** from the cable exit boot **50**.

[0090] FIGS. **7** and **8**, as discussed above, show the entire routing for all of the cabling carrying the

input and output signals within the module **12**.

[0091] It should be noted that the depicted cable routing configuration is simply one example of a configuration that can be used within the module **12** given the optical components used therein. Depending upon the type of optical components used within the module **12**, the cable management features associated with possible inserts that are configured to hold such components, or the number of signal input openings utilized, the cable routing for the module **12** can be varied.

[0092] Now referring to FIGS. **18-37**, various examples of telecommunications fixtures such as telecommunications chassis/frames that are configured to receive assemblies similar to the telecommunications assembly **10** described above are illustrated.

[0093] Referring now specifically to FIGS. **18-27**, a first embodiment of a telecommunications frame **200** that is configured to receive assemblies similar to the telecommunications assembly **10** is illustrated. In FIGS. **28-37**, a second embodiment of a frame **300** having features that are similar to the frame **200** of FIGS. **18-27** is illustrated, the frame **300** of FIGS. **28-37** providing a larger connection capacity than the frame **200** of FIGS. **18-27**. As shown, the frame **300** of FIGS. **28-37** defines a taller profile providing for a larger number of levels for receiving assemblies **10** than the frame **200** of FIGS. **18-27**.

[0094] Either frame **200**, **300** may be provided as an underground telecommunications fixture or may be provided as an above-ground fixture that can be mounted on vertical surfaces such as a telecommunications pole **400** (as illustrated in FIGS. **26-27** and **36-37**).

[0095] It should be noted that, other than the size difference, the frame **200** of FIGS. **18-27** is similar in configuration and function to that of the frame **300** of FIGS. **28-37**, and, unless specifically distinguished, the features of the frame **200** of FIGS. **18-27** are fully applicable to the features of the frame **300** of FIGS. **28-37**. Thus, the inventive features of the present application will be described with reference to only the frame **200** of FIGS. **18-27**.

[0096] Referring now specifically to FIGS. **18-27**, the frame **200** defines a cable seal portion **202** toward the bottom thereof that defines a plurality of openings **204** for sealably receiving large bundle cables **206**. It should be noted that the bundle cables **206** entering the frame **200** may include both feeder (i.e., input) and distribution (i.e., output) cables **208** carrying the respective signals. In the depicted embodiment, three openings **204** are illustrated for receiving three bundle cables **206** carrying a plurality of input and/or output individual cables **208**. Other number of openings **204** can be utilized depending upon the desired connectivity.

[0097] As will be discussed in further detail below, the outer perimeter of the cable seal portion **202** is configured to intermate with a cover/dome **210**, and, with the use of a clamp **212**, form an enclosure **214**. The cover **210** is illustrated in FIGS. **24-27**.

[0098] A module mounting portion **216** of the frame **200** extends upwardly from the cable seal portion **202** and defines a housing **218** that is configured to receive assemblies similar to the telecommunications assembly **10** discussed above, wherein such assemblies are formed from a telecommunications module similar to module **12** that has been mounted within a tray similar to tray **14**.

[0099] In FIGS. **18-20**, the module mounting portion **216** of the frame **200** is shown populated with a plurality of assemblies **10**. In FIGS. **21-23**, the frame **200** is shown in an empty configuration for illustrating the interior details of the housing **218**.

[0100] The housing **218** defines an open front **220** that is closed by a hinged door **222**. The housing **218** further defines right and left sidewalls **224**, **226** and a rear wall **228**, configured to cooperatively receive the inserted assemblies **10** through the open front **220**.

[0101] In the embodiment shown in FIGS. **18-27**, a swell latch **230** is used to keep the door **222** closed with respect to the housing **218**. Other types of latches can also be used.

[0102] Referring now to FIGS. **18** and **20**, the door **222** and the housing **218** define intermating structures for temporarily keeping the door **222** in an open position so that a technician can freely access the connection locations provided on the assemblies **10** within the housing **218**. In the

embodiment shown, the door **222** defines a slide guide opening **232** at the bottom thereof that has a narrower slide portion **234** and a wider retention portion **236**. The guide opening **232** cooperates with a guide pin **238** that is spring-loaded to be biased downwardly. The guide pin **238** extends from the housing **218** via a pivotally disposed arm **240**. The guide pin **238** defines a narrow portion **242** that is sized to slide within the narrow portion **234** of the guide opening **232**. The guide pin **238** also defines a wider portion **244** that is sized to fit within the wider retention portion **236** of the guide opening **232** but not in the narrower slide portion **234**. When the door **222** is in a predetermined position (may be the fully or partially opened position), the guide pin **238** is biased downwardly such that the wider portion **244** of the guide pin **238** is aligned with and able to move into the wider portion **236** of the guide opening **232**. Since the wider portion **236** of the guide pin **238** is not sized to slidably fit through the narrow portion **234** of the opening **232**, the guide pin **238** locks the door **222** in the predetermined, e.g., open, position. The guide pin **238** has to be lifted upward against a bias force in order to align the narrow portion **242** of the guide pin **238** with the narrow portion **234** of the guide opening **232** for slidably pivoting the door **222** to a closed position.

[0103] Referring back to FIGS. **18-20**, once each of the assemblies **10** is formed, the assemblies **10** are slidably loaded into the housing portion **218** of the frame **200** in a vertically stacked arrangement as shown in FIGS. **18-20**. Cantilever arms **246** provided on each tray **14** are used for fixing the trays **14** to the frame **200** with a snap-fit interlock.

[0104] The frame **200** is shown in an empty configuration in FIGS. **21-23** to illustrate the assembly mounting locations in greater detail. As shown, bent portions **248** on right and left walls **224**, **226** of the housing **218** define slide surfaces **250** that slidably guide each tray **14** into the frame **200**. The cantilever arms **246** cooperate with apertures **252** formed on the right and left walls **224**, **226** of the housing **218** to flexibly lock the assemblies **10** into place.

[0105] Now referring back to FIGS. **18-20**, provided around the exterior of the housing **218** are a number of cable management structures that are configured to guide individual cables **208** broken out from the bundle cables **206** entering the enclosure **214**. The cable management structures may be used to guide cables **208** to and/or from the assemblies **10** mounted within the housing **218**.

[0106] As shown from a rear perspective view of the chassis **200** in FIGS. **20** and **23**, the module mounting portion **216** defines a plurality of curved radius limiters **254** in a vertically stacked arrangement at the rear left corner, each one configured to receive cables **208** coming from radius limiters **256** positioned on the trays **14** of the assemblies **10**.

[0107] Still referring to FIGS. **20** and **23**, positioned toward the rear center of the chassis **200** are a plurality of spools **258** provided in a vertically stacked arrangement. The spools **258** are configured to guide cables **208** coming from and going to the curved radius limiters **254** of the frame **200** and take up slack for any cabling **208** extending between the different layers of modules **12**. Another set of cable spools **260** are positioned to the right of the spools **258** and may be used for either further cable slack management or to separate cables coming from a given module **12** and provide separate cable paths to different modules **12** positioned toward the front of the chassis **200**.

[0108] A cable management plate **262** is positioned underneath a majority of the cable spools **260**. The plate **262** defines a plurality of radiused passages **264** at the right side thereof for guiding broken out cables **208** from the bundled cable **206** into the rears of the modules **12**. As shown, the broken out cables **208** can enter the rear inputs of the modules **12** through an opening **266** provided at the rear wall **228** of the housing **218**. The plate **262** is configured to partially cover the opening **266** to protect the input cables **208**. When a bundled cable **206** sealably enters the enclosure **214**, cables **208** that are broken out may be looped (e.g., at least once) within the cable seal portion **202** of the frame **200** and lead upwardly to the rear sides of the modules **12** within the module mounting portion **216** of the frame **200**. When the cables **208** are lead upwardly, they pass through the radiused passages **264** defined by the plate **262**, enter into the housing **218** through the opening **266**, and are terminated or crimped to rear sides of the modules **12**.

[0109] Furthermore, the plate **262** can guide cables coming from the cable spools, either the right set **260** or the left set **258**, downwardly such that the cables can be routed back up around the spools **258** to the modules **12**.

[0110] When the cables are routed back toward the modules **12**, they pass through the curved radius limiters **254** located at the rear left corner of the frame **200** and through the radius limiters **256** of the associated trays **14** before being lead to a given module **12**.

[0111] It should be noted that the frame **200** is configured to allow for a variety of fiber connectivity implementations depending upon the types of modules **12** that are used on the frame **200**. As noted earlier, a bundled cable **206** can carry input and output signals (via broken out individual cables **208**) through the same bundled cable **206**. The cable management features at the rear of the frame **200** are used according to the routing involved between different layers of modules **12**.

[0112] For example, according to one implementation (similar to the connectivity arrangement illustrated in FIGS. **28-30**) the bundled cable **206** can feed a number of feeder or input signals into a number of the modules **12** mounted on the frame **200**. An output from a given module **12** can enter the input of another module **12** (e.g., a splitter module) at a different layer and be split into a plurality of signals that can be relayed to different module layers on the frame **200**. As discussed previously, the cable spools **258**, **260** at the rear of the housing **218** can be used in managing the cables routed between the different layers. An example of a module that houses a fiber optic splitter for this type of a connectivity arrangement may be similar to the module **12** described above.

[0113] In the illustrated example of FIGS. **28-30**, a “parking lot” feature is also provided on the chassis **300**. As shown, a number of the modules may be implemented as “parking lot” or connector storage modules **312** (the bottom two layers in the given example) that can be used to hold live-signal connectors until they are connected to a given module **12** for further processing.

[0114] As noted above, the cable seal portion **202** of the frame **200** can receive a cover **210** to seal the module mounting portion **216** from the environment. As shown in FIGS. **24-25**, abutting flanges **268**, **270** of the cable seal portion **202** and the cover **210** can be clamped via a clamp **212**. Further sealants such as an O-ring may also be used.

[0115] For a taller frame such as the frame **300** of FIGS. **28-37**, a pair of spacer collars **272** and multiple clamps **212** may be used to enclose the frame **300**. It should be noted that by using the spacer collars **272**, the same cover **210** can be utilized for both sizes of the frames.

[0116] As noted above, the frames **200**, **300** of the present disclosure may be provided as underground telecommunications fixtures or may be provided as above ground pole-mounted fixtures (as illustrated in FIGS. **26-27** and **36-37**).

[0117] As shown, for example, in FIGS. **26-27**, the frame may be mounted to a telecommunications pole **400** via a bracket assembly **402**. The bracket assembly **402** may include a pole mount portion **404** defined by an upper part **406** and a lower part **408** that is mounted on the pole **400** and a frame mount portion **410** that is mounted to the cable seal portion **202** of the frame **200**.

[0118] Adjacent the top end thereof, the cover **210** also defines a mount **412** that is configured to cooperate with the upper part **406** of the pole mount portion **404** of the bracket assembly **402** to secure the frame **200** to the pole **400**.

[0119] As shown, the frame mount portion **410** of the bracket assembly **402** that is attached to the cable seal portion **202** of the frame **200** defines a pair of pins **414** projecting from sidewalls **416** thereof. The pins **414** cooperate with notches **418** defined on the lower part **408** of the pole mount portion **404** of the bracket assembly **402**. The notch **418** defined on each side includes a locking portion **420** and an access portion **422**. When the pin **414** is in the locking portion **420** of the notch **418**, the enclosure **214** sits vertically upright and the cover **210** can be fastened to the upper part **406** of the pole mount portion **404** of the bracket assembly **402**. When one of the clamps **212** needs to be taken off to remove the cover **210** to access the interior of the enclosure **214**, first, a fastener used to couple the dome **210** to the upper part **406** of the pole mount portion **404** of bracket

assembly **402** is removed. Next, each pin **414** is moved into the access portion **422** of the notch **418** of the lower part **408** of the pole mount portion **404** of the bracket assembly **402**. Lifting the pin **414** out of the locking portion **420** of the notch **418** and sliding the pin **414** into the access portion **422** of the notch **418** brings the enclosure **214** to a pivoted/tilted position. At this point, the clamp **212** around the cover **210** can be loosened and the cover **210** can be removed for accessing the interior of the enclosure **214**. When the technician is finished with the enclosure **214**, the cover **210** may be clamped and the enclosure **214** can be tilted back to the upright position by sliding the pin **218** into the locking portion **420** of the notch **418**. And, at this point, the enclosure **214** can be fastened to the pole **400** at the upper part **408** of the pole mount portion **404** of the bracket assembly **402**.

[0120] As noted previously, unless otherwise stated, the above description is fully applicable to the taller-profiled frame **300** of FIGS. **28-37**.

[0121] Referring specifically to FIGS. **34-35**, it should be noted that in order to access all of the connection locations within the frame **300**, the middle clamp **212**, rather than the uppermost clamp **212**, has to be removed when removing the dome **210**. As such, when the dome **210** is removed, the upper collar **272** stays clamped to the dome **210** with the uppermost clamp **212**. Otherwise, the upper collar **272**, if left mounted to the lower collar **272**, may block or limit access to some of the connection locations of the frame **300**.

[0122] Referring to FIGS. **30** and **33**, it should also be noted that the taller-profiled frame **300** includes a longer cable management plate **362** at the rear exterior of the housing **318** for covering a longer opening **366** and defines a single cable management spool **360** above the plate **362** and a single cable management spool **360** below the plate in addition to the spools **358** provided to the left of the plate, all serving similar functionality as those discussed above with respect to frame **200**.

[0123] The above specification, examples and data provide a complete description of the manufacture and use of the disclosure. Since many embodiments of the disclosure can be made without departing from the spirit and scope of the inventive aspects, the inventive aspects resides in the claims hereinafter appended.

Claims

1. A telecommunications chassis comprising: a module mounting portion defining a height, the module mounting portion further comprising: a housing defining an open front leading to an interior and a rear wall, the housing further defining a plurality of module mounting locations provided in a vertically stacked arrangement within the interior, each configured to receive a telecommunications module through the open front, an exterior of the housing including a first column of radius limiters for guiding cables from the front of the housing toward the rear wall with bend control, the radius limiters of the first column being provided in a vertically stacked arrangement, wherein each module mounting location that is provided in the stacked arrangement is aligned with a separate associated radius limiter from the first column of radius limiters for guiding cables from the front toward the rear wall of the housing, wherein the stacked module mounting locations and the first column of radius limiters are each respectively horizontally aligned so as to define vertically stacked parallel planes, the rear wall defining an opening that has an exterior access portion communicating with an exterior of the rear wall, wherein the exterior access portion extends generally along a majority of the height of the module mounting portion of the telecommunications chassis for accessing, from the exterior of the housing, rear ends of modules to be mounted in the housing for signal input, wherein the exterior of the housing includes a plate at least partially overlapping the opening for protection of cables entering the opening.
2. The telecommunications chassis of claim 1, further comprising a cable sealing portion defining at least one cable opening configured to sealably receive a cable, wherein the module mounting

portion extends upwardly from the cable sealing portion, the telecommunications chassis further comprising a dome sealably attached to the cable sealing portion of the chassis for forming an enclosure.

3. The telecommunications chassis of claim 2, wherein the dome is removably attached to the chassis via a clamp.
4. The telecommunications chassis of claim 1, wherein the open front of the housing is closable by a door to define the interior, wherein, when the door is closed, the housing defines a generally cylindrical configuration.
5. The telecommunications chassis of claim 2, wherein the cable sealing portion defines a channel for looping of cabling entering the chassis through the at least one cable opening.
6. The telecommunications chassis of claim 1, further comprising a plurality of telecommunications modules mounted to the housing.
7. The telecommunications chassis of claim 6, wherein each of the telecommunications modules is mounted via flexible cantilever arms forming a snap-fit interlock.
8. The telecommunications chassis of claim 7, wherein at least one of the telecommunications modules is a connector-storage module configured to temporarily hold a plurality of live-signal connectors without relaying signals of the connectors.
9. The telecommunications chassis of claim 6, wherein at least one of the telecommunications modules carries a fiber optic component in the form of a fiber optic splitter for splitting input signals into a plurality of the same output signals.
10. The telecommunications chassis of claim 4, further comprising a spring-biased latch for keeping the door at a predetermined open position.
11. The telecommunications chassis of claim 4, further comprising a locking latch for keeping the door at a closed position.
12. The telecommunications chassis of claim 11, wherein the locking latch includes a swell latch.
13. The telecommunications chassis of claim 2, further comprising a removable extension collar provided between the dome and the cable sealing portion of the chassis.
14. The telecommunications chassis of claim 13, wherein the extension collar is clamped via removable clamps to both the cable sealing portion of the chassis and the dome.
15. A telecommunications chassis comprising: a module mounting portion defining a height, the module mounting portion further comprising: a housing defining an open front leading to an interior and a rear wall, the housing further defining a plurality of module mounting locations provided in a vertically stacked arrangement within the interior, each configured to receive a telecommunications module through the open front, an exterior of the housing including a first column of radius limiters for guiding cables from the front of the housing toward the rear wall with bend control, the radius limiters of the first column being provided in a vertically stacked arrangement, wherein each module mounting location that is provided in the stacked arrangement is aligned with a separate associated radius limiter from the first column of radius limiters for guiding cables from the front toward the rear wall of the housing, wherein the stacked module mounting locations and the first column of radius limiters are each respectively horizontally aligned so as to define vertically stacked parallel planes, the exterior of the housing including a second column of radius limiters that are spaced apart and generally parallel to the first column of radius limiters, the rear wall defining an opening that has an exterior access portion communicating with an exterior of the rear wall, wherein the exterior access portion extends generally along a majority of the height of the module mounting portion of the telecommunications chassis for accessing, from the exterior of the housing, rear ends of modules to be mounted in the housing for signal input.
16. The telecommunications chassis of claim 15, further comprising a cable sealing portion defining at least one cable opening configured to sealably receive a cable, wherein the module mounting portion extends upwardly from the cable sealing portion.

17. The telecommunications chassis of claim 15, wherein the open front of the housing is closable by a door to define the interior, wherein, when the door is closed, the housing defines a generally cylindrical configuration.

18. The telecommunications chassis of claim 15, further comprising a plurality of telecommunications modules mounted to the housing.

19. The telecommunications chassis of claim 18, wherein each of the telecommunications modules is mounted via flexible cantilever arms forming a snap-fit interlock.

20. The telecommunications chassis of claim 18, wherein at least one of the telecommunications modules carries a fiber optic component in the form of a fiber optic splitter for splitting input signals into a plurality of the same output signals.
